Comparative Analysis of Bandgap Reference Circuits

Using Different Amplifier Topologies

AICD Project Presentation

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Differential NMOS-Based Bandgap Reference

BGR Operating Principle

Reference Voltage Equation

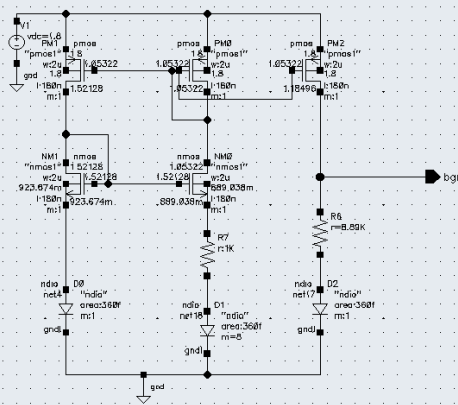
$$V_{\text{ref}} = V_{BE} + k \cdot \Delta V_{BE}$$

$$\Delta V_{BE} = V_T \ln(n), \quad V_T = \frac{kT}{q}$$

Two complementary components:

- **CTAT** (V_{BE}): Decreases with temperature $\approx -1,6\,\text{mV}/^\circ\text{C}$
- **PTAT** (ΔV_{BE}): Increases with temperature $\approx +0.085\,\text{mV}/^\circ\text{C}$

Circuit Architecture



BGR with NMOS Diff. Amp.

Folded Cascode Amplifier-Based BGR

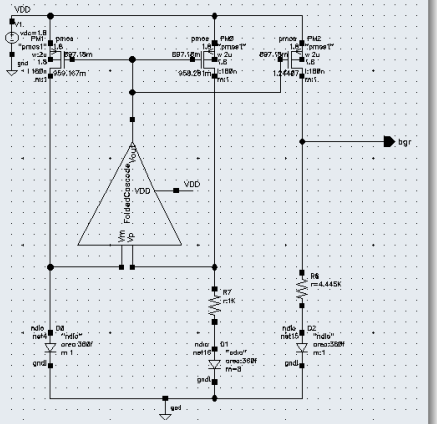
Why Folded Cascode?

- Higher open-loop gain: ≈ 80–$100\,\text{dB}$
- High output impedance $\Rightarrow$ better accuracy
- Superior PSRR due to cascoding isolation
- More accurate CTAT/PTAT cancellation

Design Considerations:

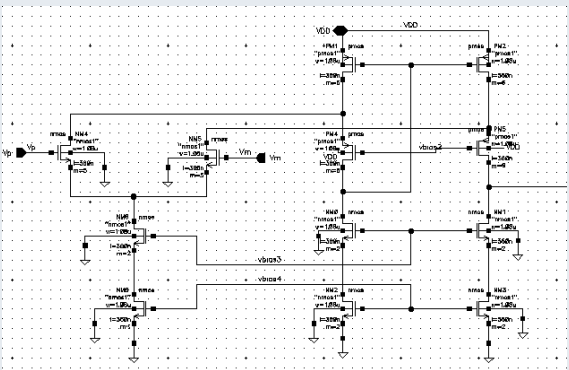
- Bias current from dedicated current mirror
- Transistors sized for saturation operation
- Resistor network (R_7, R_8) sets scaling factor k
- Diode-connected devices (D0, D1, D2) generate CTAT & PTAT

Folded Cascode BGR Circuit



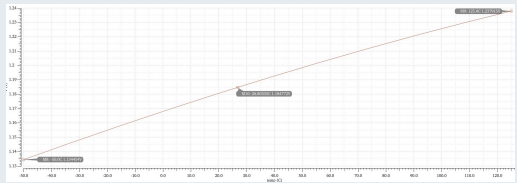
BGR with Folded Cascode Amplifier

Folded Cascode BGR Circuit

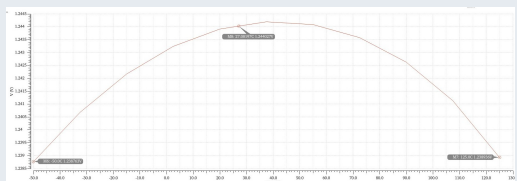


Folded Cascode

BGR

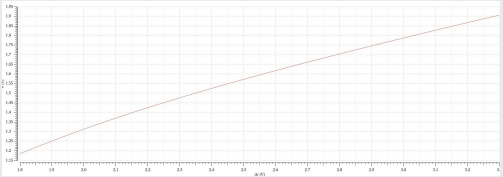


BGR with folded cascode

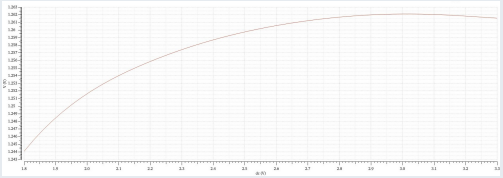


Results

Line regulation

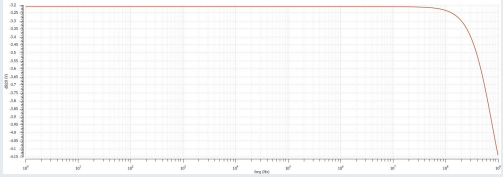


Differential

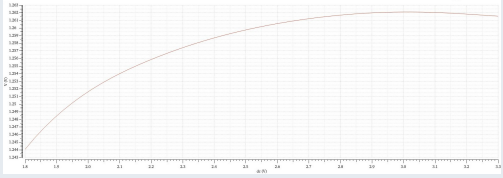


Folded cascode

PSRR



Differential



Folded cascode

Results & Comparative Analysis

Temperature Coefficient (TC)

- Differential Amp: **498** ppm/°C
- Folded Cascode: **0.80** ppm/°C

Line Regulation ($\Delta V_{ref}/\Delta V_{DD}$)

- Differential Amp: 490 mV/V
- Folded Cascode: 11.7 mV/V

Power Consumption ($P = V_{DD} \times I_{DD}$)

- Differential Amp: 149.9 μW **(lower)**
- Folded Cascode: 660.1 μW

Performance Comparison Table

Parameter	Diff. Amp	Folded Cascode
Temp. Coefficient	Higher	Lower
PSRR	Moderate	High
Line Regulation	Moderate	Better
Load Regulation	Moderate	Better
Power Consumption	Low	Moderate
Design Complexity	Simple	Complex

Thank You

"The amplifier topology is the cornerstone of BGR performance."

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